

ZHALPAQTAL, KazakhstanWMO: **343980**Lat: **49.667N**Lon: **49.483E**Elev: **33**StdP: **14.68**Time Zone: **5.00 (E05)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.4	-9.0	-19.4	1.9	-14.0	-14.2	2.5	-8.6	26.2	19.2	22.6	16.8	7.0	50	0.578

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.1	98.9	68.0	95.7	66.8	92.7	65.8	70.3	91.2	69.0	89.3	67.8	86.9	8.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.7	91.7	74.6	62.8	85.5	74.2	60.9	80.0	75.2	34.2	91.5	33.1	89.4	32.1	86.7	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	22.5	18.7	16.4	DB	-17.2	103.0	6.6	3.1	-22.0	105.3	-25.8	107.1	-29.5	108.8	-34.2	111.0
(5)				WB	-17.6	73.6	6.5	3.2	-22.2	75.9	-26.0	77.8	-29.6	79.6	-34.3	81.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	47.0	14.6	15.1	30.0	49.7	64.5	72.6	77.2	75.3	62.1	46.7	33.0	21.0	(6)
(7)		DBStd	24.52	12.47	12.27	10.77	8.61	8.40	7.90	6.19	7.55	7.81	8.43	9.72	11.63	(7)
(8)		HDD50	4396	1099	978	623	109	3	0	0	0	6	167	511	900	(8)
(9)		HDD65	7699	1564	1398	1086	462	115	23	1	12	144	569	960	1365	(9)
(10)		CDD50	3295	0	0	1	101	453	678	844	786	368	63	1	0	(10)
(11)		CDD65	1124	0	0	0	4	99	251	381	332	56	1	0	0	(11)
(12)		CDH74	14039	0	0	0	45	1246	3141	4677	4200	717	13	0	0	(12)
(13)		CDH80	6912	0	0	0	5	508	1538	2388	2221	251	1	0	0	(13)
(14)	Wind	WSAvg	8.4	9.2	9.2	8.9	9.2	8.6	8.2	7.7	7.4	7.6	7.8	8.1	8.6	(14)
(15)	Precipitation	PrecAvg	9.6	0.9	0.7	0.8	0.8	0.7	1.0	0.9	0.9	0.3	1.1	1.0	0.8	(15)
(16)		PrecMax	13.9	2.4	2.0	2.8	2.6	3.4	4.2	4.2	3.3	2.5	2.4	3.2	1.9	(16)
(17)		PrecMin	4.3	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.1	0.0	(17)
(18)		PrecStd	2.4	0.5	0.5	0.7	0.7	0.7	0.9	0.9	0.7	0.6	0.6	0.7	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.5	38.7	61.2	79.6	94.2	100.6	103.1	101.8	93.2	75.4	57.3	44.8	(19)
(20)		MCWB	38.7	36.4	49.3	62.1	63.6	67.2	69.7	69.1	64.0	55.2	51.1	42.8	(20)	
(21)		2%	DB	35.9	36.0	53.4	73.8	89.9	96.3	98.9	98.7	86.5	70.2	51.4	40.3	(21)
(22)		MCWB	34.9	34.0	44.8	56.8	63.1	66.1	68.2	68.3	62.0	53.9	47.4	38.3	(22)	
(23)		5%	DB	33.5	33.9	48.0	68.8	86.1	93.1	95.3	95.3	82.3	64.9	48.3	36.9	(23)
(24)		MCWB	32.4	32.5	42.8	54.8	63.0	65.9	67.4	66.4	60.3	52.1	45.3	35.2	(24)	
(25)		10%	DB	31.9	32.4	43.2	64.6	81.3	89.1	91.8	91.7	77.5	60.3	45.3	34.0	(25)
(26)		MCWB	31.1	31.2	38.7	52.2	61.3	64.6	66.2	65.7	58.7	50.4	42.9	32.7	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.5	36.8	50.1	63.7	67.6	71.6	73.2	71.9	67.2	60.0	53.4	43.3	(27)
(28)		MADB	40.1	38.3	59.8	77.7	85.6	91.1	90.0	93.2	86.6	67.1	56.6	44.3	(28)	
(29)		2%	WB	35.0	34.3	46.0	59.3	65.4	69.4	70.5	69.9	64.2	56.3	48.5	38.9	(29)
(30)		MADB	35.5	35.4	52.1	70.7	84.2	89.1	90.9	92.9	79.8	65.5	50.9	40.6	(30)	
(31)		5%	WB	32.7	32.8	42.8	56.2	63.7	67.6	69.2	68.4	62.0	54.0	45.6	35.6	(31)
(32)		MADB	33.5	33.8	48.4	66.0	81.9	85.8	88.6	90.2	77.4	62.9	47.8	36.6	(32)	
(33)		10%	WB	30.9	31.3	38.9	53.5	62.0	66.1	68.0	67.1	60.3	51.4	43.0	33.0	(33)
(34)		MADB	31.9	32.5	42.3	63.0	79.0	83.9	86.3	87.7	74.9	58.6	45.2	33.9	(34)	
(35)	Mean Daily Temperature Range		MDBR	11.9	13.3	14.0	19.2	21.9	22.6	22.1	23.7	22.5	18.6	11.4	10.0	(35)
(36)		5% DB	MADB	8.2	10.1	19.8	25.2	27.2	27.5	26.1	28.2	29.5	26.8	15.2	7.8	(36)
(37)			MCWBR	8.1	9.2	12.8	12.2	9.8	9.1	8.7	10.0	12.6	14.5	11.7	7.1	(37)
(38)			MADB	8.5	9.8	19.4	23.1	24.5	23.9	22.7	24.8	24.8	22.4	13.1	7.0	(38)
(39)		5% WB	MCWBR	8.5	9.2	12.7	11.7	10.2	8.8	8.2	9.7	11.5	13.8	11.0	7.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.328	0.350	0.394	0.452	0.430	0.410	0.417	0.430	0.382	0.363	0.341	0.326	(40)	
(41)		taud	2.069	2.027	2.085	2.034	2.177	2.300	2.286	2.223	2.322	2.302	2.254	2.135	(41)	
(42)		Ebn at Noon	212	240	252	250	263	270	266	255	257	238	211	195	(42)	
(43)		Edh at Noon	28	37	42	49	44	40	40	40	33	28	23	23	(43)	
(44)	All-Sky Solar Radiation	RadAvg	348	712	1077	1475	1939	2094	2030	1766	1262	733	347	235	(44)	
(45)		RadStd	80	135	151	106	140	173	136	105	110	91	52	44	(45)	